

KRISHNA SIDDHARDH POTLURI

+1 (682) 259-5852 | ksiddhardh032@gmail.com | [linkedin.com/in/ks2000](https://www.linkedin.com/in/ks2000)

EDUCATION

Arizona State University — Tempe, AZ

Aug. 2023 – May 2025

Master of Science in Computer Science

Birla Institute of Technology and Science(BITS Pilani) — Hyderabad, IN

Aug 2017 – May 2021

Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

Software Developer

Nov 2022 - Jun 2023

BeeHyv Software Solutions

Hyderabad, IN

- Developed a full-stack agricultural management platform using Django REST Framework, React, and PostgreSQL to monitor farm operations and deliver predictive insights.
- Explored CNNs for crop yield forecasting before transitioning to Transformer models in PyTorch, achieving 80% accuracy using weather data and historical yield for similar crops.
- Built Transformer-based time series models to capture long-term dependencies, enabling precise yield forecasting and better resource planning.
- Engineered autoencoder pipelines for anomaly detection in crop data, facilitating early identification of irrigation issues.
- Optimized heavy backend computations by rewriting key logic in C++, improving processing speeds by over 200% compared to Python implementations.

Project Engineer

Jul 2021 - Oct 2022

Wipro Limited

Bengaluru, IN

- Built a Python-based machine learning algorithm to detect and visualize defects in wooden electrical poles, replacing a square-based method with an innovative radial mathematics approach to actually see where the defects exactly are.
- Achieved 95% accuracy in defect detection by implementing complex radial calculations and leveraging ultrasound data for enhanced defect visualization.
- Optimized algorithm performance by 100% through data preprocessing techniques, enabling real-time processing on mobile platforms (Android/iOS) via cloud integration.
- Contributed to Flutter app development, focusing on data collection and algorithm integration, and built Flask API endpoints to seamlessly deploy the ML model for web and mobile use.
- Constructed a user friendly React-based website to enhance project accessibility and usability.
- Presented the project solo to the CTO during COVID, securing continued funding amid widespread project cuts.

Software Engineer Intern

Aug 2019 - Dec 2019

Wheebox

New Delhi, IN

- Designed and developed a full-stack web application for exam analytics, providing real-time insights on attempt rates, accuracy, and response times using Java Spring Boot APIs, MySQL, and AngularJS.
- Implemented a deep neural network (DNN) using TensorFlow and Keras to detect cheating behavior, analyzing time sequences and answer patterns with 77% classification accuracy, significantly improving exam integrity.
- Explored alternative models such as Random Forests and SVM using Scikit-learn for benchmarking, selecting DNNs for superior performance on sequential behavior data.
- Utilized Pandas and NumPy for pre-processing large-scale response data and OpenCV to track behavioral anomalies in proctored exam settings.

PROJECTS

AI-Driven Automated Pricing System for Digital Marketplaces | *Python, React, Firebase*

Present

- Engineered a scalable pricing pipeline integrating React.js frontend with Firebase backend, achieving 90.2% price prediction accuracy within \$10 margin for marketplace items through hybrid LLM-CNN architecture
- Implemented transfer learning with ResNet-50 for cost-efficient visual feature extraction, reducing computational overhead while maintaining 94.4% prediction accuracy within \$25 margin

Fake it Till you Tweet it | *Mistral AI, Python, RAG, Mixed-Regression, Research, Spark, HIPAA Awareness, Linux*

Dec. 2024

- Created a RAG-powered tweet generation system by fine-tuning Mistral models with varying prompt strengths. Used vector search for context retrieval, then evaluated susceptibility using mixed regression models, exploring ethical / clinical analogs to PHI vulnerabilities. Prototyped SMTP mail routing using Mailgun API and implemented spam detection using Naive Bayes.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, Kotlin, C/C++, JSON, SQL, NoSQL, HTML5, CSS, jQuery

Frameworks: TensorFlow, PyTorch, Pandas, Keras, Django REST, Node.js, React.js, AngularJS, Spring framework (JDBC, JPA, MVC), Android (Jetpack), Next.js, Redux, GraphQL, Express, Bootstrap, Material UI, Hibernate, Restful API, Open API

Other: Git, Machine Learning, Deep Learning, Python, Scikit-learn, Neural Networks, Natural Language Processing (NLP), Computer Vision, Reinforcement Learning, MLOps, Model Deployment, Cloud Computing (AWS), Statistical Analysis, Data Preprocessing, Feature Engineering.